On Causal Discovery Methods and Applications to Criminal Justice

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Abstract

Causal discovery aims to recover graph structures that facilitate interpretation and intervention. This is especially relevant in criminal justice, where practitioners are critically interested in understanding which factors may causally influence recidivism. This thesis studies causal discovery in this setting using the National Institute of Justice (NIJ) recidivism dataset, a public mixed-type empirical dataset. The work evaluates six causal discovery algorithms, FCI, DAGSLAM, DAGBagM, Hill Climbing, LiM, and ReX, on a synthetic benchmark dataset and the NIJ dataset. The synthetic benchmark is used to compare behaviour under known causal structures, while the empirical study examines the plausibility and robustness of learned graphs in a noisy real-world setting. Two robustness diagnostics, self-compatibility and edge-stability, are introduced to assess how consistently graph structure is recovered across repeated runs and varying subsets of variables. The results show that no single algorithm performs best across all criteria. Instead, methods differ substantially in graph density, stability, and substantive plausibility, with performance depending strongly on the assumptions imposed by each algorithm and the structure of the data. These findings support a hybrid multi-algorithm workflow in which different methods are used at different stages of the discovery process rather than treated as direct substitutes. The thesis therefore contributes an empirical comparison of causal discovery methods in criminal justice, a practical workflow for mixed-type recidivism data, and a robustness-based perspective on evaluating learned causal graphs.

Keywords: causal discovery; recidivism; criminal justice; directed acyclic graphs; mixed-type data; robustness; causal inference

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1 The Problem of Recidivism

Causal discovery is concerned not only with prediction, but with identifying structure that supports explanation and intervention. This matters especially in criminal justice, where policymakers are interested not only in who is likely to reoffend, but in which factors may causally influence risk. This thesis studies causal discovery in criminal justice, using recidivism as a concrete application domain and the NIJ dataset as the empirical test case.

Recidivism, or an individual's reoffending behaviour in spite of prior punishment [1, 2], is a pervasive, costly, and policy-relevant issue across many jurisdictions. In the United States alone, two-thirds of prisoners were rearrested for at least one new serious crime post release, with over half ending up back in prison within just 3 years [3]. Repeat offending in the world's largest economy contributes up to 20% of all adult arrests made [4]. These statistics are not geographically idiosyncratic either: recidivism rates in Denmark, Sweden, and Finland were greater than 30% in the 2018 cohort taken with a 2 year follow up period, and Latvia, Taiwan, and Australia have all separately reported cohorts with a 50%+ reoffending rate [5].

The social cost of recidivism is substantial. At a micro level, reoffending may reflect a failure of rehabilitation efforts and a return to risk-compounding circumstances. At a macro level, high levels of recidivism place strain on already pressured prison systems [6], raise questions about the efficacy of intervention programmes, and create opportunity costs for other areas of public spending such as infrastructure, education, and healthcare. This has led to decades of research and the development of operationalised risk assessment and classification tools aimed at targeting individuals most at risk of reoffending, with desistance serving as a central policy objective. Yet this approach remains limited in scope, particularly in its ability to identify the concrete causal levers through which reoffending might be reduced.

This work asks two questions: (RQ1) how do causal discovery algorithms behave on the NIJ recidivism dataset, a real-world mixed-type criminal justice dataset? (RQ2) Can a hybrid multi-algorithm approach produce more robust and reliable causal graphs than any single method alone? The thesis makes three contributions: an empirical study of six algorithms (FCI, DAGSLAM, DAGBagM, Hill Climbing, LiM, and ReX) on synthetic benchmarks and the NIJ dataset; a hybrid workflow guided by dataset characteristics; and a robustness evaluation using self-compatibility and edge-stability. Supporting code is available at `https://github.com/wigglypint/On-Causal-Discovery-Methods-and-Applications-to-Criminal-Justice`.

1.1 The Motivation for Causal Models

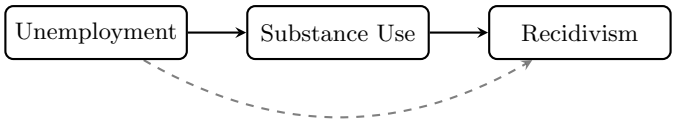


Figure 1: A hypothetical causal graph illustrating a mediation outcome confounder: *Substance Use* mediates the relationship between *Unemployment* and *Recidivism*. The dashed edge represents a weak or negligible direct effect of unemployment on recidivism once substance use is accounted for. A causal model encoding this structure identifies substance use programmes as the higher-leverage intervention.

A fundamental limitation of actuarial approaches to recidivism risk goes back to the adage "correlation is not causation". Policy makers are ultimately interested in what interventions are practical, insofar as managing causal antecedents of recidivism, something that causal models explicitly encode: "identifying causal dynamic risk factors for violence is crucial for developing focused interventions to reduce risk" [7]. The better the causal graph, the more appropriately resources can be directed towards monitoring and intervention.

To illustrate, consider a hypothetical causal graph in Figure 1 over recidivism risk factors in which *Substance Use* mediates the relationship between *Unemployment* and *Recidivism*, such that the direct effect of unemployment on reoffending is negligible once substance use is accounted for. A purely predictive model would flag both unemployment and substance use as risk factors of similar importance; a causal model reveals that substance use programmes are the higher-leverage intervention, with employment schemes of

secondary importance. This is the kind of reasoning a causal graph enables, and that purely association-based approaches cannot. Whether causal discovery algorithms can reliably recover such structures from real criminal justice data, and which algorithms are best suited to do so, is the central question of this work.

2 Related Work

Niu et al. (2024) [8] showed that the performance of causal discovery algorithms depends strongly on domain characteristics. Their work benchmarked 29 algorithms across synthetic and applied datasets varying in linearity, noise structure, and sample size. The present work takes a similar practitioner-oriented perspective, but focuses specifically on recidivism and criminal justice data.

Prior work in criminology and sociology has identified predictive factors for recidivism such as criminal history, social environment, and substance use, many of which underpin modern risk assessment tools. However, these approaches remain predictive rather than causal. Applying causal discovery in this domain is especially challenging since datasets in the domain are mixed-type, noisy, and likely confounded, all conditions known to degrade performance [8]. To our knowledge, this is among the first studies to compare multiple causal discovery algorithms systematically on criminal justice data.

Causal analysis has also been used to study fairness in algorithmic decision making by asking whether protected attributes have causal effects on decisions [9]. This is especially relevant in criminal justice, where work on COMPAS has argued that racial bias is embedded in predictive systems [10, 11]. Beyond concerns of fairness, causal methods have also been used to evaluate anti-recidivism policies, for example in studies of ex-combatants in Colombia [12]. This use of causal methods in the pursuit of improving policy evaluation within criminal justice only reinforces the the symbiotic link between the two fields.

3 Recidivism: Theories and Literature

3.1 Risk Assessment Models through Time

Risk assessment methods have evolved from unstructured clinical judgement to actuarial tools and, later, to models incorporating dynamic risk factors. Early clinical approaches suffered from low inter-rater reliability, while actuarial tools improved consistency but relied heavily on static historical variables and offered little guidance on how risk might be changed [13, 14]. Later generations of tools introduced criminogenic needs and other dynamic factors, reflecting a shift from prediction alone towards intervention-focused assessment [14].

3.2 Risk-Need-Responsivity Model

The Risk-Need-Responsivity (RNR) model, introduced by Andrews, Bonta et al. in 1990 [15], became foundational for third and fourth-generation risk assessment tools. The model is organised around three core principles: intervention intensity should reflect the individual's risk of reoffending (the risk principle); treatment should target criminogenic needs (the need principle); and programmes should be responsive to the individual's capacities and learning style (the responsivity principle) [16, 15]. Although the model has been criticised for de-emphasising non-criminogenic factors, review evidence suggests that programmes targeting criminogenic needs are more effective at reducing recidivism than other methods [17]. For the present study, the key point is that criminogenic needs are dynamic risk factors directly linked to criminal behaviour and therefore natural candidates for causal variables in recidivism [14].

3.3 Dynamic causal factors in literature

Criminogenics is a widely used term in literature to describe attributes and characteristics that lead an individual to commit a crime. They are therefore an excellent starting point for expert hypotheses on variables which may be candidate causal variables. Note that criminogenic needs refer to fluid characteristics which, when changed in some way, can bring about a reduction in reoffending [18]. These are therefore also attractive targets for policy makers designing rehabilitative interventions.

The concept of causality is well known in the allied fields of criminology and psychology - Kraemer et al (1997) (as cited in pg.351 Douglas & Skeem, 2005) [7] identifies causal dynamic risk factors as a variable which is shown to:

1. precede and increase the likelihood of violence;
2. change spontaneously or through intervention; and
3. predict changes in the likelihood of recidivism (violence) when altered

Some common criminogenic factors referenced in literature are summarised below.

Table 1: Criminogenic factors commonly appearing in literature

Criminogenic Factor	Description	RNR
Anti-social Peer Associations [16, 19, 14, 7]	This encompasses pro-criminal social support and a lack of "anticriminal others" [19, 20].	✓

Anti-social Personality Pattern [16, 19, 14, 20]	As per *The Diagnostic and Statistical Manual of Mental Disorders 5E* [21], this includes traits of but is not limited to: 1. Impairments in self functioning: Encompassing **Identity** and **Self-direction**. 2. Impairments in interpersonal function: Principally issues with **Empathy**, and **Intimacy**. 3. Antagonism: As distinguished by **Manipulativeness**, **Deceitfulness**, **Callousness**, and **Hostility**. 4. Disinhibition: Identified by **Irresponsibility**, **Impulsivity**, and **Risk taking**.	✓
Substance Abuse [16, 22, 19, 23, 14, 7, 20]	Including alcohol and other drugs.	✓
Anti-social Cognition [19, 14, 7, 20]	Anti-social cognition refers to attitudes, thought processes, and beliefs which lead to engagement in anti-social behaviours. For example, anger, defiance, or identification with a criminal identity.	✓
(Lack of) Accommodation [22, 24, 20]	This is inclusive of temporary accommodation and rough sleeping.	
(Lack of) Education / Employment [22, 19, 25, 14, 20]	Identified by poor scores in measures of achievement and fulfilment at school/work.	✓
Familial and/or Marital [19, 14, 20, 26]	Familial and marital bonds contribute in two key ways [19]. They provide a sense of belonging and care, as well as being natural sources of supervision.	✓
Leisure and/or Recreation [19, 14, 26]	Characterised by a lack of engagement or fulfilment derived from non-criminal pastimes.	✓
Finances/Income [27, 28, 13]	This captures financial difficulties.	
Mental Disorders [20, 29, 30, 28]	As distinct from just personality related disorders, including clinical conditions such as schizophrenia.	

Repeat offenders tend to exhibit more criminogenic traits than non-repeat offenders, and interventions targeting a greater number of criminogenic needs are associated with stronger reductions in recidivism [18, 22, 16]. This provides further evidence for treating these factors as candidates for causal investigation.

3.4 Static Risk Factors

Aside from criminogenic factors, static risk factors are favoured for their tendency to be quantifiable and commonly collected. Although these are not variables of interest from an interventional perspective, they are nontheless potentially relevant causal variables.

Table 2: Static risk factors affecting recidivism

Static Risk Factor	Description	RNR
Criminal History [18, 28, 20, 31]	Used to identify a lifespan-long pattern of offending.	✓
Age [18, 20, 31, 13]	Recidivism is negatively correlated with age.	
Traumatic Experiences [20]	Adverse psychological or physical trauma experienced in an individual's history.	

The factors marked with a ✓ in the RNR column correspond to the "Central Eight" of Andrews and Bonta [32]. These are often grouped into the "Big Four" (Criminal History, Anti-social Personality Pattern, Anti-social Cognition, Anti-social Peer Associations) and the "Moderate Four" (Familial/Marital, Education/Employment, Leisure/Recreation, Substance Abuse).

3.5 Risk Assessment Tools

Risk assessment instruments provide another perspective on expert hypotheses, since they encode the factors practitioners themselves consider relevant to reoffending. These tools are commonly divided into actuarial and structured professional judgement approaches, but both are useful here primarily as sources of variable selection rather than as causal models in their own right [33]. We summarise a selection of widely used instruments below.

Table 3: Selected risk assessment tools used for recidivism prediction and management

Instrument	Abbreviation	Authors	Country	SPJ?
Correctional Offender Management Profiling for Alternative Sanctions	COMPAS	Northpointe (1996) [29]	USA	✓
Offender Assessment System	OASys	Howard (2006) [27]	UK	✓
Level of Service Inventory-Revised / Case Management Inventory	LSI-R / LS/CMI	Andrews & Bonta (1995) [30], Andrews, Bonta, & Wormith [34, 26]	UK, NZ, CA, USA	✗
Offender Group Reconviction Scale	OGRS3	Howard (2009) [35]	E&W	✗
Recidivism Risk Assessment Scales	RISc	Montfoort Consultancy & Dutch Probation Service (2004) [36]	NL	✗
Post-Conviction Risk Assessment	PCRA	Johnson et al. (2011) [13]	USA	✗
Ohio Risk Assessment System	ORAS	Latessa et al. (2009) [28], Ohio Department of Rehabilitation & Correction [37]	USA	✗
Historical Clinical Risk Management–20	HCR-20 V3	Douglas et al. (2014) [20]	Global	✓
Static-99	Static-99	Harris e al. (2003) [31]	Global	✗
Violence Risk Appraisal Guide	VRAG	See Harris, Rice, & Quinsey (1993), Quinsey (2019) [38]	CA, UK	✗
Violence Risk Scale	VRS	Lewis, Olver, & Wong (2013) [39]	CA, UK, AU	✗

Appendix B provides a fuller mapping of these instruments to the Central Eight factors identified by Andrews and Bonta [32].

3.6 Rehabilitation Program Reform

Rehabilitation-focused interventions provide a complementary view of potentially causal drivers of recidivism, since successful programmes often target factors believed to be modifiable and related to offending. Examples such as the Boston Reentry Initiative, evaluations of Ohio halfway houses, and community-based correctional facilities emphasise support in areas including employment, housing, substance use, mental health, and education [40, 41]. Although these programmes differ in design and population, they reinforce the relevance of these variables as candidate causal drivers.

3.7 NIJ Challenge: Winning Models

The 2021 NIJ Recidivism Forecasting Challenge produced several winning predictive models and accompanying analyses of influential variables. The NIJ synthesis report [42] shows that variable importance changes across the first, second, and third years post release, but several themes recur across submissions: age at release, criminal history, gang affiliation, employment stability, drug-testing, and supervision-related indicators are all cited. These variables should not be interpreted causally, and the report itself notes that predictive importance does not reveal the direction or mechanism of influence [42]. Nonetheless, because causal relationships often co-occur with strong associations [43], the challenge results provide useful empirical evidence for a list of plausible variables.

4 Notation and Definitions

Before turning to the algorithms themselves, it is useful to introduce the core notation and concepts that causal discovery relies on. Unlike predictive modelling, causal discovery is not only concerned with patterns of association in data, but with whether those patterns can be interpreted as evidence for underlying structural relationships. Assumption This makes the assumptions behind each method, and the way their outputs are evaluated, particularly important. In mixed-type, noisy criminal justice data, these assumptions are rarely satisfied perfectly, and understanding them is essential for interpreting both the strengths and limitations of the algorithms examined in this thesis.

Definition 4.1 (Directed, Partially Directed, & Undirected Graphs)**.** A graph $G = (V, E)$ is a tuple comprising a set of nodes V and a set of edges $E \subseteq V \times V$. An *undirected* edge $e = \{u, v\} \in E$ is an unordered pair of vertices indicating that u and v are connected. In contrast, a *directed* edge $e = (u, v) \in E$ is an ordered pair indicating a connection oriented from u to v, commonly denoted $u \to v$. A directed graph has edge set E comprising solely directed edges, and similarly an undirected graph has edge set E comprising solely undirected edges. Finally, a partially directed graph has an edge set containing both directed and undirected edges.

For the sake of simplicity, the rest of the definitions will assume directed graphs.

Definition 4.2 (Directed Acyclic Graphs (DAG) & Causal Graphs)**.** A *directed acyclic graph* (DAG) is a directed graph $G = (V, E)$ containing no directed cycles. In the causal discovery setting, a *causal graph* is a DAG in which nodes V represent random variables and directed edges $(u, v) \in E$ encode the assertion that u is a direct cause of v, relative to the variable set V.

Definition 4.3 (Adjacency)**.** Let $G = (V, E)$ be a graph, and let $u, v \in V$. We say that u is *adjacent* to v if $(u, v) \in E$ or $(v, u) \in E$. We denote the set of vertices adjacent to u in the graph G by

$$adj(u, G) := \{v \in V : (u, v) \in E \text{ or } (v, u) \in E\}.$$

Definition 4.4 (Cycles & Acyclicity)**.** A cycle $C = v_1 e_1 \dots e_n v_{n+1}$ is a sequence of vertices and edges such that $\forall\ e_i \in C,\ e_i = (v_i, v_{i+1})$ and $v_{n+1} = v_1,\ v_i \neq v_j\ \forall\ i, j \notin \{1, n+1\}$.

Definition 4.5 (Parent)**.** Let $G = (V, E)$ be a directed graph, and let $v \in V$. A node $p \in V$ is said to be a *parent* of v if there exists $e \in E$ such that $e = (p, v)$. We denote by Parents(v) the set of all parents of the node v, and extend this definition to sets of nodes $X \subseteq V$ by

$$\text{Parents}(X) = \bigcup_{v \in X} \text{Parents}(v).$$

Definition 4.6 (Descendant)**.** Let $G = (V, E)$ be a directed graph, and let $v \in V$. A vertex $d \in V$ is said to be a descendant of v if there exists a directed path $P = v \to v_2 \to \cdots \to d$ for some length $|P| \geq 1$. We denote by Descendants(v) the set of all descendants, and extend this definition to sets of nodes $X \subseteq V$ by

$$\text{Descendants}(X) = \bigcup_{v \in X} \text{Descendants}(v).$$

Definition 4.7 (Markov Equivalence Class [44])**.** Let G_1, G_2 be DAGs with the same node set V. G_1 and G_2 are said to be Markov equivalent iff they share the same skeleton and set of v-structures $(X \to Z \leftarrow Y)$.

Definition 4.8 (Partial Ancestral Graph)**.** A partial ancestral graph (PAG) $\mathcal{P}$ over node set V is a mixed graph representing the Markov equivalence class of maximal ancestral graphs consistent with some set of conditional independences. A graph is a member of the Markov equivalence class if it is consistent with the PAG. Edges in a PAG end in either $-, \to, \circ$, where $\circ$ denotes that graphs in the Markov equivalence class may have either a tail or an arrow in this position.

The graph-theoretic definitions above describe the types of objects causal discovery algorithms attempt to recover. To interpret their outputs meaningfully, however, it is also necessary to understand the assumptions under which such recovery is possible. These assumptions are rarely perfectly satisfied in practice, especially in observational criminal justice data, but they provide a useful framework for assessing when an algorithm's conclusions can be treated with relative confidence, and when they should be interpreted more cautiously.

4.1 Common Properties and Assumptions Made by Causal Discovery Algorithms

Definition 4.9 (Causal Markov Condition [44, 45])**.** Let G be a graph over a node set V, and $P(V)$ be a probability distribution. $P(\cdot)$ is said to satisfy the Markov condition *iff* for every $W \in V, W \perp\!\!\!\perp V \setminus \text{Descendants}(W) \cup \text{Parents}(W) | \text{Parents}(W)$. That is, given the set of all its parents, any vertex in G is probabilistically independent of all non-descendents.

Definition 4.10 (Faithfulness Condition [44, 45])**.** We say that the Causal Markov Condition *entails* others when **every** probability distribution having the independence relations implied by the Markov condition has those further independence relations.

Let G be a graph and P a probability distribution on G. P and G are said to be *faithful iff* no conditional independence holds unless entailed by the Causal Markov Condition.

Definition 4.11 (Causal Sufficiency [44, 45])**.** Let S be a set of variables. S is *causally sufficient iff* every common cause of any two (or more) variables in S is also in S.

In more common terms, this assumption means that we assume there are no latent confounders.

Definition 4.12 (Confounder)**.** A variable C is said to be a *confounder* of two variables X and Y if C is a common cause of both X and Y, i.e. $C \to X$ and $C \to Y$ in the causal graph. A *latent confounder* is a confounder which is not observed or quantified in the dataset.

In addition to their underlying assumptions, causal discovery algorithms must also be evaluated in different ways depending on setting. In synthetic benchmarks, performance can be measured directly against a known ground-truth graph. In empirical settings, where no such graph is available, robustness and internal consistency are our barometers.

4.2 Metrics for the Evaluation of Causal Discovery Algorithms

Definition 4.13 (Structural Hamming Distance)**.** Let G_1, G_2 be graphs. $\text{SHD}(G_1, G_2)$ computes the difference between the adjacency matrices of G_1, G_2. That is, the minimal number of transformations (insertions and deletions of edges) that must be made to G_2 before it is identical to G_1. A lower structural hamming distance between the learned causal graph and the ground truth is therefore desirable.

Definition 4.14 (Incompatibility Score [46])**.**

$$\kappa^G(\mathcal{A}, X) := \frac{1}{k} \sum_{i \in [k]} \text{SHD}(\mathcal{A}(X), \mathcal{A}(X_{S_i}))$$

Where SHD is structural hamming distance, and $\mathcal{A}(X)$ is the causal graph learned on the variable set X. X denotes the full set of observed variables, while X_{S_i} denotes the restriction to a subset $S_i \subseteq X$ of size k.

Definition 4.15 (Precision [8])**.**

$$\textit{Precision} = \frac{\textit{true positives}}{\textit{true positives} \ + \ \textit{false positives}}$$

Definition 4.16 (Recall [8])**.**

$$\textit{Recall} = \frac{\textit{true positives}}{\textit{true positives} \ + \ \textit{false negatives}}$$

This is sometimes referred to as power.

Definition 4.17 (Robustness)**.** In this context, we use *robustness* to refer to the stability of a recovered causal graph under perturbations of the input data. An algorithm is considered robust if small changes to the dataset, such as subsampling variables or changing random seeds, produce graphs that are structurally similar (i.e. in SHD terms). Robustness is operationalised in this work through the self-compatibility and edge-stability metrics defined above.

5 Causal Discovery Algorithms

All causal discovery algorithms share a common objective: to recover a causal graph consistent with the structure of the observed data. This is an inherently ill-formed problem, since in general, many graphs may be consistent with the same set of conditional independences, and without additional assumptions, the true causal structure is unidentifiable from observational data alone. The algorithms studied in this work differ in the assumptions they impose to make the problem tractable, the search strategies they employ, and the class of graph they return. The portfolio studied here was chosen following the evaluation framework of Niu et al. [8] and with an emphasis on methods that are either compatible with mixed data or useful as practical baselines. Table 4 summarises their key assumptions.

5.1 Fast Causal Inference (FCI)

Fast Causal Inference [44, 47] developed from the PC algorithm [44], a foundational constraint-based method which recovers causal structure under the assumption of causal sufficiency. FCI relaxes this assumption, permitting latent confounders in the data generating process. Starting from a fully connected undirected graph, the algorithm iteratively removes edges when variables become conditionally independent given some set of variables. The algorithm then attempts to orient as many edges as possible via a set of orientation rules, outputting a partial ancestral graph (PAG) which corresponds to the Markov equivalence class of DAGs consistent with the conditional independences. In our experimental setup, FCI operates on the observed data using the *Fisher-z test* with a p-value threshold of $\alpha = 0.05$.

5.2 DAGSLAM

NOTEARS is an optimisation algorithm which seeks to optimise a score over the space of weighted d-variable adjacency matrices $W \in \mathbb{R}^{d \times d}$ subject to a differentiable constraint $h(\cdot)$ enforcing acyclicity: $h(W) = 0$ (Zheng et al. 2019 [48]). This has been found by Niu et al. [8] to achieve low false positive rates for non-linear datasets with both Gaussian and non-Gaussian noise. In our application, we seek a similar optimisation based approach which can natively handle mixed types, and so we employ DAGSLAM [49], an extension of NOTEARS which implements a mixed type likelihood function capable of handling both continuous and categorical data. More concretely, for the full data matrix $\mathbf{X}$ and weighted adjacency matrix W, a continuous variable $\mathbf{X}_j$ contributes to the loss function as:

$$||\mathbf{X}_j - w_j^T \mathbf{X}||_2^2 \tag{1}$$

or the usual least-squares regression loss. Similarly, for a binary variable X_k, the contribution takes the form:

$$-\langle \mathbf{X}_k, w_k^T \mathbf{X} \rangle + log(1 + exp(w_k^T \mathbf{X})) \tag{2}$$

Overall, the loss function is then the sum of losses of each variable:

$$L(W) = \sum_{j \in C} L_j + \sum_{k \in D} L_k \tag{3}$$

where C, D are the sets containing indices of continuous and discrete variables respectively.

5.3 DAGBagM

Similarly to DAGSLAM, DAGBagM utilises a composite scoring method to treat continuous and binary nodes in their natural domains. Continuous nodes are modelled via conditional Gaussian distributions while binary nodes use logistic regressions.

In particular, continuous nodes X are scored by regressing the set X onto its parents.

$$\text{score}_{\text{BIC}_X} := \frac{\sum_{i=1} (y_i - f(x_i))^2}{n} + |pa_X^G| log(n) \tag{4}$$

Binary nodes, denoted here as Z, are scored by regressing Z onto its parents via logistic regression.

$$\text{score}_{\text{BIC}_Z} := -2\left(\sum_{k=1}^{n} I(Z_k = 1)log(\hat{p}_k) + \sum_{k=1}^{n} I(Z_k = 0)log(1 - \hat{p}_k)\right) + |pa_Z^G|log(n) \tag{5}$$

Here, $\hat{p}_k = P(Z_k = 1|pa_{Z,k}^G)$, and n is the sample size. The score of the overall graph G is given by summing each variable's score:

$$\text{score}(G, Y) = \sum_{X \in C} \text{score}_X + \sum_{Z \in D} \text{score}_Y \tag{6}$$

The method repeatedly resamples data, learns a high-scoring DAG over each such resample using hill-climbing, and then aggregates such DAGs into a consensus graph (bootstrap aggregation). Edges which are frequent and increase a global score are retained, while anomalous edges are discarded. This strategy aims to reduce false positives and improve robustness of the algorithm to small perturbations in data.

5.4 Hill Climbing Algorithm (HC)

Hill climbing is included as a simple and widely used score-based baseline, implemented here through the *bnlearn* package [50] with the BIC score and no structural priors. The method greedily searches over single-edge additions, deletions, and reversals, accepting score-improving moves that preserve acyclicity until no local improvement remains. Although simple, it provides a useful reference point against which to compare the more specialised mixed-data methods.

The algorithm itself makes no particular assumptions, being a simple greedy search, and thus the implicit assumptions are based on the use of the Bayesian network model.

5.5 Linear Mixed Data (LiM)

Linear Mixed Data (or LiM) [51] is an algorithm for mixed data types assuming a linear structural equation model with non-Gaussian noise term. Niu et al. recommends DirectLiNGAM for linear and non-Gaussian datasets, while LiM is an extension of this model family to mixed types and is therefore included in testing. The algorithm entails a two part search: the first part which globally optimises a log-likelihood score function characterising the joint distribution of mixed data, and the second part, which searches combinatorially over single-edge perturbations of the skeleton in order to escape local optima.

5.6 ReX

ReX is included as a flexible machine-learning-style comparator for noisy, potentially non-linear data [52, 53]. The method repeatedly predicts each variable from all others using black-box regressors, then uses SHAP values to identify candidate parents and aggregates these across runs before thresholding weak or inconsistent edges. Its role in the portfolio is to test whether a highly flexible predictive approach offers practical advantages over more classical score- and constraint-based methods in the NIJ setting.

5.7 Summary

A summary of the portfolio causal discovery algorithms, along with their assumptions under which correctness holds, and their implementation library is provided below.

Table 4: Assumptions made by each causal discovery algorithm evaluated in this work. Faithfulness: assumes the faithfulness condition (Definition 4.10). Sufficiency: assumes causal sufficiency, i.e. no latent confounders (Definition 4.11). CMC: assumes the Causal Markov Condition (Definition 4.9). Linear: assumes linear relationships between variables.

Causal Discovery Algorithms

Algorithm	Faithfulness	Sufficiency	CMC	Linear	Software
FCI	✓		✓		causal-learn
DAGSLAM	✓	✓	✓	✓	pgmpy
LiM	✓	✓	✓	✓	LiNGAM
DAGBagM	✓	✓	✓		dagbagm (R)
HC	✓	✓	✓	✓	bnlearn (R)
ReX	✓		✓		CausalExplain

Many of these algorithms assume causal sufficiency 4.11, which is likely too strong of an assumption. Care must therefore be taken when evaluating the causal structures they learn, noting likely unmeasured confounding and exercising domain knowledge where appropriate.

One algorithm that was initially considered for inclusion in the portfolio but ultimately excluded is ProDAG [54], a recent Bayesian variational inference approach to DAG learning. This was originally intended as a complement to the point-estimate approaches of other algorithms such as DAGSLAM and HC. Its exclusion was driven by practical constraints: implementation complexity, non-native mixed type compatibility, combined with the required experiments spanning multiple seed and subset sizes rendered its inclusion infeasible within the project timeline. Evaluating such methods represents a natural extension of the work taken here, particularly given its demonstrated superiority reported in the original work.

6 Methodology

The original project specification described collaboration with Unilink, a UK-based provider of rehabilitation services, with a view to incorporating proprietary offender data into the empirical component of this work. Following initial discussions, it was realised that access to this data would be infeasible within the timeframe and ethical clearance constraints of an undergraduate project. As a result, we rely on the publicly available NIJ Recidivism Challenge dataset, which provides a sufficiently large empirical setting for this study.

6.1 Libraries

The collection of packages containing implementations of the causal discovery methods used are: *bnlearn [50], pgmpy [55], causal-learn [56], LiNGAM [57], and CausalExplain [53].* No tuning of hyper parameters was undertaken, and algorithms were all used in their "vanilla" form.

6.2 CausalPitfalls [58]

The CausalPitfalls dataset, produced by Du et al. (2025) as a benchmark for causal reasoning, provides synthetic datasets sampled from ground-truth causal structures corresponding to common statistical pitfalls. Each dataset contains over 500 samples. These scenarios are used as controlled benchmark settings for comparing causal discovery algorithms before moving to the noisier empirical NIJ setting.

The benchmark covers ten scenario types, each defined by a distinct underlying causal structure. Because the datasets were originally designed for LLM evaluation, they are accompanied by contextual labels and semantically meaningful node names. For this study, however, these labels are not substantively important: of relevance is the underlying causal graph associated with each pitfall scenario. A summary of the tested scenarios is given below, while the full set of ground-truth graphs and dataset counts used in our experiments are provided in Appendices C and D respectively.

The number of synthetic datasets evaluated varied by scenario because of practical constraints encountered during experimentation. In a small number of cases, one or more algorithms failed to return a valid output for a given dataset. These counts are also reported in Appendix D.

FCI is the only algorithm considered here that returns a PAG rather than a DAG. To compute TPR, FPR, and SHD against the known ground truths, we therefore evaluate FCI on the skeleton of its returned PAG, that is, the undirected adjacency matrix obtained by treating any mark between two nodes as an edge.

1. Treatment mediation (`treatment_mediator`)
2. Causal effect (`causal_effect`)
3. Counterfactual reasoning (`counterfactual_reasoning`)
4. Causal direction (`causal_direction_iv`)
5. Setting agnosticism (`domain_shift`)
6. Confounding mediators (`mediation_outcome_confounder`)
7. Treatment effect modifiers (`moderation_effect`)
8. Markov equivalences (`dag_structure_markequiv`)
9. Distinguishing necessity and sufficiency (`necessity_sufficiency`)
10. Sequences of mediators (`sequential_mediator`)

6.3 NIJ Recidivism Dataset [59]

We now discuss the challenges presented by the NIJ dataset and the methods adopted to mitigate these.

6.3.1 Pre-processing

The dataset had missing values across 11 variables in total. A full summary metadata is detailed in Appendix A. Notably, `Gang_Affiliated` information was missing for all individuals for whom `Gender` was female (comprising $\sim 12.3\%$ of individuals), and intuitively important variables such as `Prison_Offense` were missing for around 12.7% of the dataset. To handle this missingness, we employed the `Synthetic Data Vault` [60] package available in Python to perform conditional sampling on the missing variable conditioned on all known variables less the target recidivism flags and other previously imputed variables. To limit the effects of using synthetically sampled data which exposes the data to model error deriving from the imputation, imputation was used only on a select set of variables, opting to drop the other rows corresponding to variables with a small number of missing values. The set of variables we sampled for comprised: `Gang_Affiliated, Avg_Days_per_DrugTest, DrugTests_THC_Positive, DrugTests_Cocaine_Positive, DrugTests_Meth_Positive, DrugTests_Other_Positive,` and `Prison_Offense`. After the imputation, we dropped $6,419$ rows, leaving $19,416$ entries, a retention rate of $\sim 75.2\%$. We created flag variables indicating the imputation of missing data, for example `Gang_Affiliated_Missing` indicates that `Gang_Affiliated` had been imputed for that entry.

Next, we flattened categorical variables via one-hot encoding. For this dataset, one-hot encoding led to a large increase in the number of variables. To mitigate the effects of this, we regrouped categories, sacrificing some resolution to make the problem tractable. Concretely, this involved "re-bucketing" expanded categorical variables. For example: `Prior_Arrest_Episodes_Felony_3, ..., Prior_Arrest_Episodes_Felony_10-or more` were all combined into one variable `Prior_Arrest_Episodes_Felony_high`. Even then, the number of variables posed a tractability problem despite relatively powerful compute: 32 CPUs with 96GB of RAM. We therefore further cut variables to produce a "lean" subset. This lean set was chosen to align closely with criminological frameworks such as the Central Eight risk factors.

Table 5: "Lean" NIJ variable subset, grouped by domain and mapped to the Central Eight risk factors.

Domain	**Variables in Subset**	**Mapping to Central Eight Risk Factors**
Demographics	`Gender_F; Race_BLACK; Age_at_Release;`	While gender, age, and race are not criminogenic needs, they feature in many actuarial tools as baseline correlates and capture exogenous societal structures which might cause recidivism through mediating variables.
Criminal history & antisocial pattern	`Prior_Arrest_Episodes_Felony_high; Prior_Arrest_Episodes_Property_high; Prior_Arrest_Episodes_Drug_high; Prior_Conviction_Episodes_Felony_high; Gang_Affiliated;`	Aligns with "criminal history" component of Central Eight and encodes offending patterns as well as frequency and severity of offences. Gang affiliation captures antisocial peer associations, a key criminogenic need reflecting pro-criminal social networks and lack of pro-social support.
Employment & education	`Percent_Days_Employed;`, `Jobs_Per_Year;`	Employment stability maps directly to the education and employment domain of the Central Eight; unstable employment and a lack of education is associated with higher recidivism risk and provide potential targets for intervention.
Substance use	`Avg_Days_per_DrugTest;`, `DrugTests_Cocaine_Positive;`	Positive cocaine tests serve as a proxy for issues with substance use, mapping to the Central Eight domain of the same name. This criminogenic need is strongly linked to recidivism and often a target of intervention.

Domain	Variables in Subset	Mapping to Central Eight Risk Factors
Supervision & compliance	`Delinquency_Reports_high;`, `Supervision_Risk_Score_First;`	Delinquency reports characterise compliance with authority, while supervision risk score aggregates multiple dynamic and static factors; both link to anti-social cognition and personality patterns.
Outcome	`Recidivism_Within_3years;`	Target outcome against which criminogenic needs are theorised and empirically linked.

The NIJ dataset included a column for race, only specifying White or Black. This was one-hot encoded into a binary indicator for race (Black vs non-Black (i.e White)). We retain this variable in the DAGs as a static attribute, facilitating exploration of possible routes by which structural inequalities (e.g. differential policing, sentencing, employment opportunities) may influence the risk of recidivism. We reiterate that we explicitly do not interpret race as a biological cause of offending. Instead, we read outgoing edges as proxies for upstream social processes.

6.3.2 Post-hoc Processing

A post-processing pipeline was developed to obtain interpretable causal graphs from the algorithm's raw output. Firstly, we harmonised age representation used in the dataset. The one-hot encoded dataset originally contained seven mutually exclusive age buckets. These dummy variables were collapsed into a single node `Age_at_Release` to improve graph legibility. More concretely, all age-bucket nodes were treated as manifestations of a latent age variable, and so defined an edge between `Age_at_Release` and some other variable `X` whenever any age-bucket node was adjacent to `X` in the learned DAG. The edge was oriented via majority direction across buckets, and ties were discarded. This first step leaves the fitted model unchanged, but allows for a sparser and more interpretable representation of any effects arising due to age.

Secondly, to enforce the proper chronology, we projected each learned DAG onto a local subgraph comprising only the target node `Recidivism_Within_3years` and its direct parents. This is justified because the target variable is an event occurring after supervision start. Thus, only variables measured at or before realisation of the target variable could plausibly be considered candidate causes. This projection is algorithm-agnostic, respects temporal ordering, and highlights a small set of variables that the discovery procedure proposes as direct causes of recidivism.

6.3.3 Self-compatibility

The self-compatibility experiments were split into two distinct threads. One tested for meaningful variation across three random seeds `[42, 7, 12]`, computing the incompatibility score using 50 randomly sampled subsets of variables where each subset satisfied $|S| = 5$, drawn from the set of lean variables. This tests whether an algorithm learns consistent structures when presented with partials of the same data. The second thread tested the sensitivity of algorithms to the size of the variable subset with the fixed seed 42. We tested the subset sizes $k =$ `[5,10,15]`. As proposed by Faller et al. [46], self-compatibility score can be used as a proxy for structural hamming distance in the absence of ground truth, algorithms with a high self-compatibility score should therefore be viewed with caution when used in this domain.

6.3.4 Edge-stability

Edge stability experiments were conducted to characterise the ability of algorithms to recover edges consistently when run on perturbations of data. We ran $B = 20$ bootstrap resamples of the lean NIJ dataset, sampling with replacement a new dataset equal in size to the original. The output for each resample is a learned DAG (post-hoc processed as described in Section 6.3.2). We then recorded the frequency each candidate edge appeared across B resamples. We call an edge stable if it appears in at least 16 of the 20 resamples. This illuminates a different aspect of algorithmic performance, keeping the variable set static and

perturbing data. Due to the computational cost of 20 full resamples, the experiment was only conducted for DAGBagM, DAGSLAM, and HC, with results reported in Table 11.

7 Results

This section reports results for two settings: a synthetic benchmark using the CausalPitfalls datasets, and an empirical analysis of the NIJ recidivism dataset. Full experimental methodology is given in Section 6. Synthetic datasets contain over 500 samples each; counts varied by scenario due to algorithmic failure on a subset of runs (Appendix D). The NIJ dataset contained 19,416 entries after preprocessing (about 75.2% retention) and was reduced to a lean subset for tractability on available hardware (32 CPUs, 96GB RAM). Self-compatibility experiments used seeds `[42, 7, 12]` and 50 random subsets of size $|S| = 5$; edge-stability experiments used the same seeds across $k \in \{5, 10, 15\}$.

7.1 Synthetic Benchmark

We begin with the synthetic benchmark to compare algorithmic behaviour in a controlled setting where ground-truth structure is known. Figures 2 and 3 should be read together: the violin plots summarise the overall distribution of performance across runs, while the scatter plots make the trade-off between true and false edge recovery more explicit at the level of individual runs.

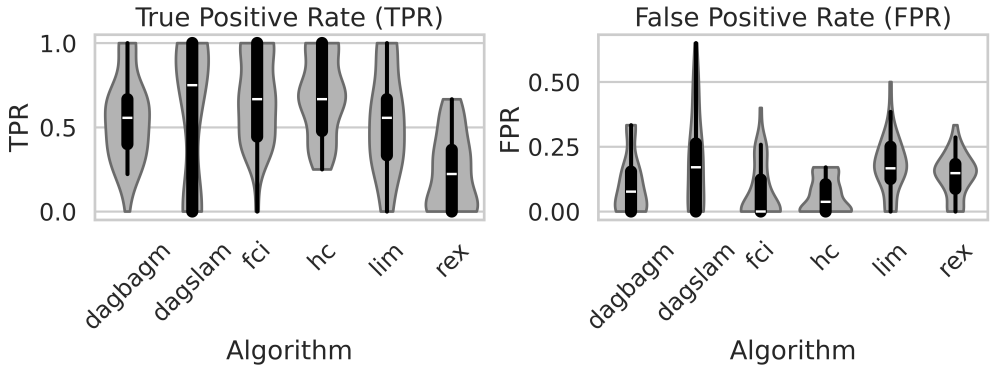


Figure 2: Violin plots of true positive rate (TPR) and false positive rate (FPR) on the CausalPitfalls benchmark, grouped by algorithm. These distributions show not only average behaviour but also the variability of each method across benchmark scenarios, allowing comparison of how reliably different algorithms recover true edges while avoiding false discoveries.

We can make several observations from these runs. Strikingly, ReX behaves conservatively, with a noticeably lower-centred TPR distribution than the other algorithms. From the scatter plot we see clear clustering around the bottom left region, suggesting a tendency to miss true edges for the sake of limiting false discoveries. By contrast, DAGSLAM often achieves high TPR values at the cost of a wider and heavier-tailed FPR distribution. Its scatter plots therefore suggest a more aggressive edge-recovery strategy, one that can improve sensitivity while also increasing the risk of overfitting. Hill Climbing appears comparatively balanced in this benchmark. Its TPR and FPR distributions are more stable than those of the more extreme methods, and its scatter points are more consistently concentrated toward the top-left region. This suggests that, at least in the synthetic setting, HC offers a stronger compromise between recovering true structure and avoiding false positives than either the more conservative ReX or the more aggressive DAGSLAM.

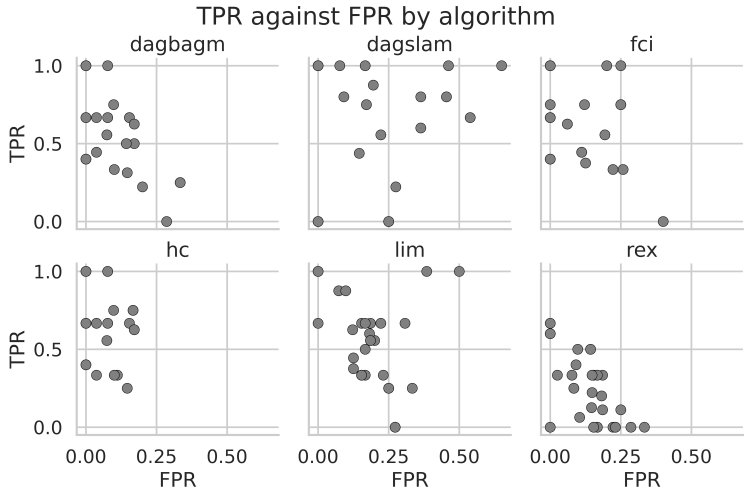
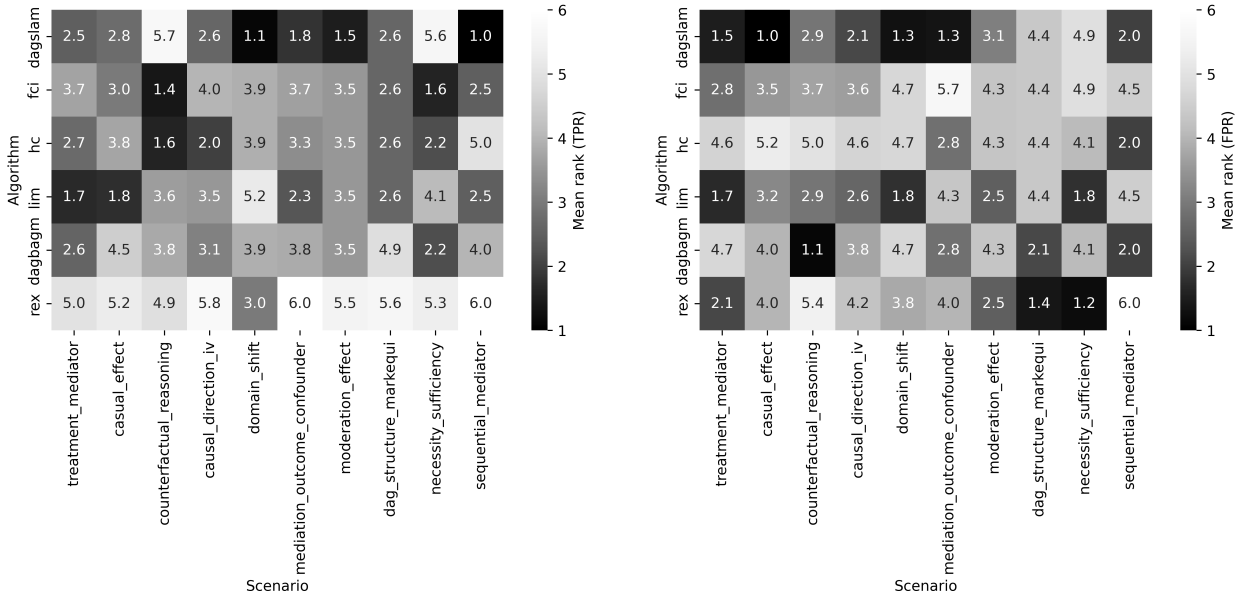


Figure 3: Scatter plots of TPR against FPR for each algorithm on the CausalPitfalls benchmark. The key feature to note is where points cluster: points closer to the top-left indicate stronger performance, corresponding to high recovery of true edges with fewer false positives, while broader scatters suggests less stable behaviour across scenarios.



(a) Ranking by TPR

(b) Ranking by FPR

Figure 4: Scenario-level algorithm rankings by mean true positive rate (TPR) and false positive rate (FPR), averaged across datasets within each CausalPitfalls setting. These heatmaps are intended to highlight where particular algorithms perform relatively well or poorly depending on the underlying causal structure, rather than to identify a single universally best method.

Figure 4 shows that relative algorithm performance varies meaningfully by causal scenario. Rather than pointing to a single dominant method, the benchmark suggests that different algorithms are better suited to different structural tasks, a motivator for a hybrid approach.

In the treatment-mediator setting, the data is generated so that the intervention affects the target outcome through an intermediate variable rather than solely through a direct path. LiM achieves the best mean rank

on a TPR basis by a substantial margin (approximately 0.8 rank points better than the next algorithm), implying that it most reliably recovers the treatment mediator outcome pathway. Conversely, DAGSLAM ranks best on FPR, with LiM close behind (about 0.2 rank points worse), suggesting DAGSLAM produces marginally sparser graphs with fewer false positives. In a recidivism setting, where policymakers may wish to understand how interventions reduce reoffending via intermediate criminogenic needs, this pattern suggests using a high-TPR method such as LiM for exploratory recovery, while treating a lower-FPR method such as DAGSLAM as a useful cross-check on edge credibility.

In the causal-effect setting, represented by a DAG with multiple independent treatment-mediator pathways, LiM again attains the best mean TPR rank. It performs 1.0 rank point better than DAGSLAM and 2.0 ranks better than hill climbing on average, indicating stronger recovery of true structure even when part of the effect is mediated via third variables. This corroborates the result from the treatment-mediator setting. However, FPR ranks DAGSLAM first by a clear margin, with LiM trailing by 2.2 ranks on average. This suggests that DAGSLAM is considerably more conservative when proposing edges, particularly among ultimately independent causal variables.

Of particular practical interest is the problem of correctly orienting causal edges, since the efficacy of policy interventions often depends on this. Here the ground truth contains both a mediation path and an independent direct cause pointing towards a common outcome, making orientation especially challenging. TPR highlights that hill climbing was noticeably more successful at characterising true edges correctly than both DAGSLAM and LiM, by 0.6 and 1.5 ranks respectively. On the other hand, the FPR ranks show that DAGSLAM and LiM tend to introduce fewer relative false positives, while hill climbing performs worst on this metric on average. These patterns motivate a possible heuristic for practitioners: first use DAGSLAM or LiM to obtain a relatively conservative skeleton with fewer spurious edges, and then, where edges are present, apply hill climbing to refine orientation. This strategy is guided by, but not guaranteed by, the benchmark results.

7.2 Empirical Treatment

The empirical study applies `DAGBagM`, `DAGSLAM`, `LiM`, `HC`, and `ReX` to the lean subset of the processed NIJ dataset described in Table 5. The target variable of interest is `Recidivism_within_3years`. The aim is twofold: first, to recover plausible candidate parents of 3-year recidivism in a realistic criminal justice setting; and second, to assess how robust the learned structures are under empirical noise using the self-compatibility and edge-stability diagnostics introduced in Section 6.

FCI, described in Section 5.1, is excluded from the empirical runs. Unlike the other methods, FCI returns a PAG rather than a DAG, so its output is not directly comparable in this setting without an additional conversion procedure. Since the empirical analysis is intended to compare algorithm outputs on a common basis, we restrict attention here to methods that return DAGs directly.

Table 6 summarises the high-level characteristics of the learned graphs after the post-hoc processing described in Section 6. In particular, it shows substantial variation in both graph size and runtime across methods. This is already suggestive of different algorithmic behaviours: some methods produce richer but denser hypotheses about the empirical structure, while others return much sparser graphs.

The graphs reported in this section should be interpreted as algorithm-specific causal hypotheses rather than as recovered ground truth. Since no true empirical graph is known for the NIJ data, comparisons between methods are necessarily relative: they are assessed with respect to criminological theory, internal coherence, and the robustness diagnostics reported later in this chapter.

Table 6: High-level summary of the DAGs learned by each algorithm on the NIJ dataset after post-hoc processing. The key quantities to note are the large differences in runtime and graph density, which give an early indication of how aggressively or conservatively each method behaves in the empirical setting.

Algorithm	Assumes Linearity	Assumes Suff.	Runtime (mins)	# nodes	# edges
DAGBagM		✓	6.07	9	29
DAGSLAM	✓	✓	10.9	3	3
ReX			72.8	2	1
HC	✓	✓	0.02	4	4
LiM	✓	✓	48.2	2	1

7.2.1 DAGBagM on NIJ

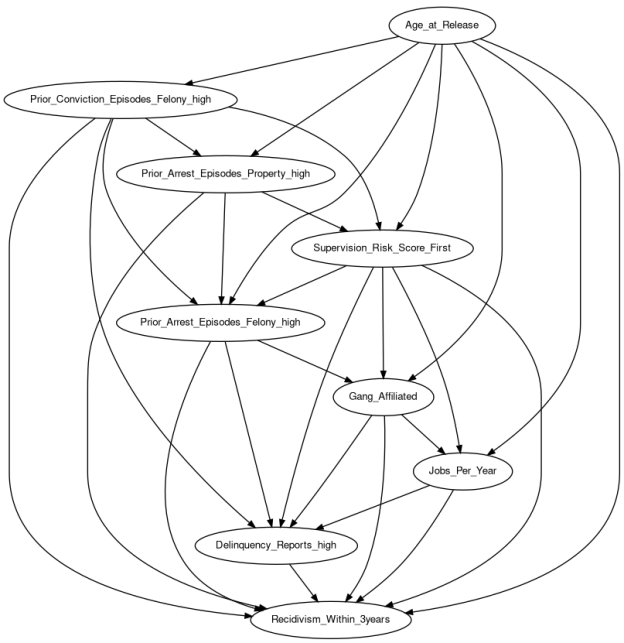


Figure 5: Pruned DAG learned by DAGBagM on the NIJ dataset, restricted to the parents of `Recidivism_Within_3years` and with age buckets merged into a single node. The main feature to note is the relative density of the recovered parent set, which includes criminal history, supervision behaviour, employment, gang affiliation, and substance-use related variables.

Figure 5 shows the DAG learned by DAGBagM, restricted to the parents of `Recidivism_Within_3years`, and with all age buckets merged into one variable. The recovered structure is dense compared to results from other algorithms, with direct parents including criminal history (`Prior_Arrest_Episodes_Felony_high;`, property and drug arrests), supervision behaviour (`Delinquency_Reports_high;`, `Supervision_Risk_Score_First;`), gang affiliation, employment (`Percent_Days_Employed;`, `Jobs_Per_Year;`), and substance use indicators (`Avg_Days_per_DrugTest;`, `DrugTests_Cocaine_Positive;`). Substantively, this is appealing because many of these variables align with criminogenic needs identified in the Central Eight framework, making the recovered graph relatively interpretable from a policy perspective. At the same time, the graph is notably denser than those returned by the other methods, so interpretability alone is not sufficient to treat all recovered edges as equally credible. Nevertheless, earlier benchmark results did not suggest that DAGBagM was uniquely prone to indiscriminate edge recovery, and its contrast with the far sparser DAGSLAM graph suggests that the density here reflects a genuinely different empirical bias rather than simple over-enthusiasm.

7.2.2 DAGSLAM on NIJ

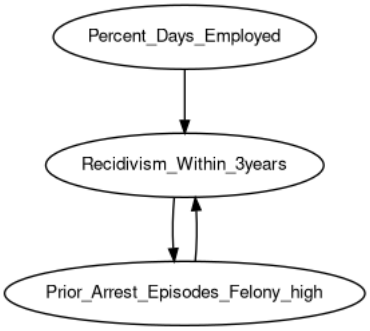


Figure 6: Pruned DAG learned by DAGSLAM on the NIJ dataset, restricted to the parents of `Recidivism_Within_3years` and with age buckets merged into a single node. In contrast to DAGBagM, the graph is highly sparse, with only a very small number of direct parents retained.

The graph produced by DAGSLAM is noticeably sparser (Figure 6) than that produced by DAGBagM. Only two direct parents of recidivism are recovered: `Percent_Days_Employed` and `Prior_Arrest_Episodes_Felony-_high`. Both are plausible from a criminological perspective, but the omission of other theoretically relevant factors, including gang affiliation, substance use, and supervision behaviour, suggests that DAGSLAM is acting conservatively in this empirical setting.

This is particularly interesting when set against the synthetic benchmark, where DAGSLAM often traded lower false-negative behaviour for a heavier false-positive tail. In the NIJ setting, however, the method appears to move in the opposite direction, returning a sparse and selective graph. One interpretation is that the mixed-type empirical data and post-hoc processing steps interact with DAGSLAM in a way that suppresses edge recovery more strongly than in the benchmark setting.

7.2.3 ReX on NIJ

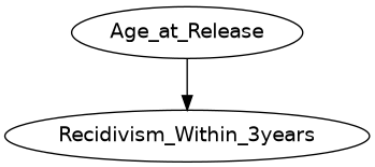


Figure 7: Pruned DAG learned by ReX, restricted to the parents of `Recidivism_Within_3years` and with age buckets merged into a single node. Notice how the structure is extremely sparse, recovering a minimal parent set

ReX behaves in line with what was observed in the synthetic cases with relatively few edges being drawn. Although age is a well-established statistic in literature, the absence of other strongly theoretically motivated links suggest that ReX is under-sensitive in this noisy setting. In other words, its sparsity does not appear to reflect a carefully selective recovery of the most credible structure, but rather a failure to recover much structure at all.

This interpretation is supported by its self-compatibility scores reported in Tables 7-10. ReX records some of the weakest robustness behaviour despite recovering so few edges, implying that even its limited output is not especially stable. In this setting, out of the box ReX therefore appears to have limited value as a stand-alone empirical tool.

7.2.4 HC on NIJ

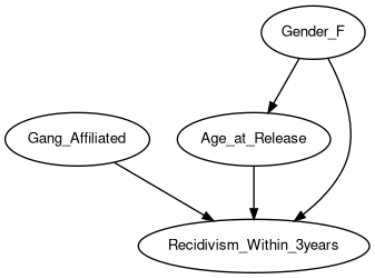


Figure 8: Pruned DAG learned by HC, showing parents of target variable with merged age node

Hill-climbing produces a richer but nonetheless compact parent set for recidivism (Figure 8). The pruned graph proposes three parents: `Age_at_Release, Gang_Affiliated,` and `Gender_F`. Age and gang affiliation are both familiar from criminological theory, whereas the direct edge from gender to recidivism is more difficult to interpret as a meaningful criminogenic need. Compared with DAGBagM and DAGSLAM, HC does not recover employment or criminal-history variables, suggesting that its greedy score-based search settles on a smaller and more selective set of associations, though not necessarily the most theoretically satisfying one.

7.2.5 LiM on NIJ

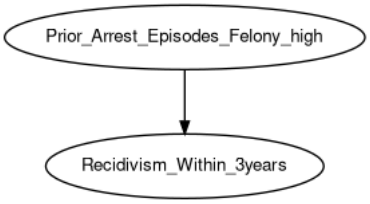


Figure 9: Pruned DAG learned by LiM on the NIJ dataset, restricted to the parents of `Recidivism_Within_3years` and with age buckets merged into a single node. The graph is highly sparse, recovering only a single direct parent.

LiM yields one of the sparsest graphs of all, linking only `Prior_Arrest_Episodes` to the target variable. This edge is consistent with the view that criminal history is a strong predictor of recidivism but does not link to any of the other criminogenic domains posited by current theory. We attribute this to the linear SEM assumption showing itself to be too restrictive. On the other hand, the surviving edge can be viewed with relatively high confidence, in that it survives the linearity assumption made.

The empirical graphs already suggest substantial variation in how aggressively different algorithms recover candidate parents of recidivism. However, visual inspection alone is not enough: a graph that appears plausible in a single run may still be unstable under perturbations in sampling. For this reason, we next examine self-compatibility, which measures how consistently each algorithm recovers edge structure across repeated random subset experiments.

7.2.6 Self-compatibility

Self-compatibility was evaluated across random seeds `[42,7,12]` using 50 samples of 5-variable subsets. ReX was limited to 10 samples for tractability reasons. Tables 7 and 8 report the resulting disagreement scores in absolute and percentage terms. Lower values indicate greater consistency across runs, and therefore more stable recovery of graph structure.

Table 7: Self-compatibility scores across random seeds. Lower values indicate greater consistency in the edge sets recovered across repeated random subset experiments, and therefore more stable graph recovery.

Algorithm	**s = 42**	**s = 7**	**s = 12**	σ
DAGBagM	7.44	8.28	6.78	0.61
DAGSLAM	3.00	3.54	3.22	0.22
ReX ($\star$)	7.10	6.00	6.20	0.48
HC	4.90	5.26	5.68	0.32
LiM	0.94	1.20	1.44	0.20

($\star$) ReX is run with 10 samples

Table 8: % of edges disagreeing across random seeds. This normalised view complements Table 7 by showing the proportion of unstable edges relative to each algorithm's recovered graph size.

Algorithm	**s = 42**	**s = 7**	**s = 12**	σ
DAGBagM	37.2%	41.4%	33.9%	3.07
DAGSLAM	15.0%	17.7%	16.1%	1.11
ReX ($\star$)	35.5%	30.0%	31.0%	2.39
HC	24.5%	26.3%	28.4%	1.59
LiM	4.7%	6.0%	7.20%	1.02

($\star$) ReX is run with 10 samples

The self-compatibility results show substantial variation in robustness across algorithms. LiM is by far the most stable method on this measure, with the lowest disagreement scores in both absolute and percentage terms. DAGSLAM also performs relatively well, with moderate but consistently low disagreement across seeds. By contrast, DAGBagM and ReX show markedly poorer stability, despite their very different graph densities: DAGBagM recovers a rich parent set but does so inconsistently, while ReX remains unstable even though it proposes very few edges. HC occupies an intermediate position. These results are important because they qualify the graph-level interpretations above: plausibility alone is not sufficient if the recovered structure is not reproduced consistently across runs.

Table 9: Self-compatibility scores when varying subset size k. Lower values indicate greater consistency in the edge sets recovered across experiments, and therefore more stable graph recovery.

Algorithm	**k = 5**	**k = 10**	**k = 15**	σ
DAGBagM	7.44	28.62	52.48	18.40
DAGSLAM	3.00	10.84	15.46	5.14
ReX ($\star$)	6.70	26.6	57.8	21.0
HC	4.90	17.0	28.2	9.52
LiM	0.94	5.60	16.28	6.42

($\star$) ReX is run with 10 samples

Table 10: % of edges disagreeing when varying subset size k. This normalised view complements Table 9 by showing the proportion of unstable edges relative to each algorithm's recovered graph size.

Algorithm	**k = 5**	**k = 10**	**k = 15**	σ
DAGBagM	37.2%	31.8%	25.0%	4.99
DAGSLAM	15.0%	12.0%	7.36%	3.14
ReX ($\star$)	33.5%	29.6%	27.5%	2.49
HC	24.5%	18.9%	13.4%	4.52
LiM	4.70%	6.22%	6.42%	1.25

($\star$) ReX is run with 10 samples

The self-compatibility experiments find that LiM exhibits the lowest incompatibility score across random seeds ($\mu \approx 1.19, \sigma = 0.20$), which corresponds to $\sim$ 6.0% of edges on subsets of 5 variables randomly sampled from the lean variable space disagreeing with the originally recovered graph on average. However, this stability is at least partly an artefact of sparsity: because LiM typically proposes only a single edge, there is relatively little structure available to disagree with the original graph. Its low incompatibility should therefore not be read straightforwardly as evidence of superior empirical reliability.

In contrast to this, DAGBagM exhibits the highest incompatibility scores.This likely reflects the fact that denser recovered graphs expose more edges to disagreement across repeated subset experiments. In that sense, the metric penalises richness in much the same way that it rewards sparsity. ReX also performs poorly despite yielding a graph sparse graph. This is notable, because it suggests that ReX is not simply abstaining from edge recovery in a stable way; rather, even its limited output appears inconsistent across runs. Although ReX was evaluated on only 10 subsets for tractability reasons, the relatively modest standard deviations across seeds and subset sizes provide little evidence that substantially more runs would qualitatively change this picture.

DAGSLAM and HC occupy an intermediate position. DAGSLAM shows relatively low incompatibility across both seeds and subset sizes, and its percentage disagreement falls as subset size increases. HC is less stable than DAGSLAM but remains noticeably more consistent than DAGBagM. These results suggest that the denser empirical graphs should be treated more cautiously, while sparse methods may appear robust for reasons that are partly mechanical. Self-compatibility is therefore informative, but only when interpreted jointly with graph density and plausibility.

7.2.7 Edge-stability

Edge-stability experiments were run only on the three more promising empirical methods, partly for tractability reasons, since each algorithm had to be refit on the NIJ dataset for $B = 20$ resamples. Whereas self-compatibility measures agreement at the level of whole recovered graphs, edge stability focuses more directly on which specific edges recur across repeated runs. This gives a more targeted indication of which candidate causal relationships are robust enough to take seriously in the absence of known empirical ground truth.

Table 11: Edge-stability results on the lean NIJ dataset across 20 resamples. The table reports how many edges recur in at least 50% and 80% of runs, together with the parents of recidivism that remain stable under the stricter 80% threshold.

Algorithm	# edges (freq $\geq$ 0.5)	# edges (freq $\geq$ 0.8)	Stable ($\star$) parents of recidivism
DAGBagM	128 (43.0%)	80 (26.8%)	`[Age_at_Release, Supervision_Risk_Score_First, Prior_Arrest_Episodes_Property_high, Prior_Conviction_Episodes_Felony_high, Prior_Arrest_Episodes_Felony_high]`
DAGSLAM	88 (75.2%)	72 (61.5%)	`[Percent_Days_Employed]`
HC	75 (49.0%)	56 (36.6%)	`[Gender_F, Gang_Affiliated]`

($\star$) We take stable to mean that an edge exists between the parent and the target variable in more than 80% of runs

Table 11 shows that DAGSLAM is the most stable of the three methods at the edge level. A large majority

of its recovered edges recur in at least 50% of runs, and over 60% survive even the stricter 80% threshold. However, this stability comes with a trade-off: the only highly stable parent of recidivism recovered by DAGSLAM is `Percent_Days_Employed`. The method therefore appears robust, but also highly selective.

DAGBagM presents almost the opposite profile. It recovers a much richer empirical structure, and several parents of recidivism remain stable at the 80% level, including age, supervision risk, and prior property- and felony-related history. This is substantively attractive, since these variables are more easily reconciled with criminological theory. At the same time, the proportion of stable edges is much lower than for DAGSLAM, reinforcing the earlier conclusion that DAGBagM provides a richer but less stable picture of the empirical graph.

HC again occupies an intermediate position. Its stable parent set is small, and the persistence of `Gender_F` alongside `Gang_Affiliated` illustrates both the promise and the limitation of the method: it recovers some intuitively plausible structure, but also stabilises at least one edge that is harder to justify as a meaningful criminogenic mechanism.

Overall, the edge-stability results support the view that no single algorithm simultaneously maximises richness, stability, and theoretical plausibility. This provides direct empirical motivation for the hybrid workflow discussed in the next chapter.

8 Evaluation & Discussion

This chapter returns to the two research questions posed in the introduction. RQ1 asked how state-of-the-art causal discovery algorithms behave on a mixed-type criminal justice dataset such as NIJ. RQ2 asked whether a hybrid multi-algorithm workflow can produce more robust and reliable causal graphs than any single method alone.

8.1 Synthetic benchmark interpretation

The synthetic benchmark helps answer the first research question, namely how different causal discovery algorithms behave across settings that isolate specific causal challenges. The results reveal a careful balance between two complementary modes of failure: algorithms that propose many edges recover ground truths well but incur penalties from high false positive rates, while relatively conservative algorithms prioritise avoiding spurious edges at the cost of recovering true edges. No single algorithm clearly dominates across all scenarios, which already suggests that a hybrid workflow may be better suited to the full causal discovery task than reliance on any one method alone.

8.1.1 LiM

LiM achieves the best mean TPR rank across the portfolio algorithms in the treatment mediation and causal effect scenarios (Figure 4a). This shows that the algorithm is well equipped to recover the true edge set when the mechanism is linear, and that the skeleton search implemented is effective in escaping local optima. As expected, the FPR rank of LiM is weaker, presenting the algorithm as a high-recall, low-precision algorithm in the linear case.

8.1.2 DAGSLAM

DAGSLAM consistently ranked highly (first or second) by mean FPR rank across all causal pitfall scenarios at the expense of TPR. This behaviour is the inverse of LiM's, trading sensitivity for accuracy. Practically, this suggests DAGSLAM may be more useful as a filtering step for thinning dense candidate graphs than as a standalone recovery method.

8.1.3 HC

It is of interest to note that hill climbing outperformed both DAGSLAM and LiM by a significant margin (0.6 and 1.5 ranks on average, respectively) in the causal direction scenario. This is consistent with the algorithm design which exhaustively permutes local edges and selects the model with lower loss. Combined with its poor mean FPR rank on the same experimental set, we surmise that where edges were not spurious, hill climbing was well equipped to orient them in the correct direction, despite proposing spurious edges in addition. This suggests that hill climbing may be useful as an edge orientation "device" which can be combined with a low-sensitivity, high-accuracy algorithm to validate edge directions.

8.1.4 ReX

ReX exhibited noticeably conservative behaviour in all synthetic scenarios, consistently clustering towards the bottom left of the TPR-FPR plot (Figure 3). This carried over into the empirical setting, recovering one of the sparsest graphs amongst all algorithms. Crucially, ReX was by far the most computationally expensive algorithm (Table 6), with a runtime of 72.8 minutes on the lean NIJ subset, 7x the runtime of LiM. This suggests that with a relatively small variable set, the computational cost of an ML-based approach does not yield a richer or obviously better causal structure. This is supported by the self-compatibility score (Table 9), which did not indicate better stability as a result of any specificity.

8.1.5 Implications for the Empirical Setting

The synthetic benchmark also begins to answer the second research question, namely whether a hybrid, multi-algorithm workflow can produce more reliable causal structures than any single method alone. The

benchmark results caution against selecting one algorithm as a universally applicable structure learner for recidivism data. Any underlying causal mechanism for recidivism is almost certainly non-linear and contains latent confounders due to practical constraints, attributes which variously violate linearity and causal sufficiency assumptions. Nonetheless, the benchmark results help motivate a practical workflow for interpreting the empirical results: edges appearing in both LiM and DAGSLAM graphs are given the most credence and are then compared to the orientation of the same edges, where they exist, in the HC graph.

We note that this workflow relies on LiM being able to recover a sufficiently dense candidate graph. This is a reliance which we revisit when applying the workflow to the empirical data in Section 8.2.

8.2 Empirical Robustness & Stability

8.2.1 Algorithmic behaviour in the noisy setting

The empirical results help complete the answer to the first research question by showing how differently the algorithms behave when applied to a real, noisy, mixed-type criminal justice dataset. A wide gap in recovered graph densities emerges across methods: DAGBagM recovers the densest graph with 29 edges, while LiM recovers the sparsest graph, just 1 edge (Table 6). This is most likely reflective of fundamental differences in modelling assumptions made by each algorithm. We saw that in the synthetic case with data drawn from a model with linear SEMs, LiM consistently topped rankings against other algorithms by TPR (Figure 4a). If relationships in the data are non-linear, then the assumptions made by LiM on the underlying are misspecified, causing the algorithm to be unable to detect edges. In contrast to this, DAGBagM uses bootstrap aggregation to piece together a causal graph, making no linearity assumptions on the underlying model, and is correspondingly able to recover a structure that is much richer. This gap in recovered graph densities suggests that the linearity assumption made by LiM is too restrictive in this domain.

The self-compatibility tests offer a way to understand how robust algorithms are in the absence of ground truth. Examining the scores in Tables 7 and 9, DAGSLAM exhibits the most consistent scores across variations in random seed and subset sizes. This suggests that DAGSLAM recovers a sparse structure regardless of initialisation, lending credibility to its edge abstention. DAGBagM and ReX both exhibited high incompatibility scores, proposing substantially different structures depending on the subset of variables. And, as discussed in Section 7.2.6, LiM's exceptionally low incompatibility score is understood to be an artefact of extreme sparsity.

8.2.2 Revisiting the proposed workflow

As noted in Section 8.1.5, the proposed two-algorithm workflow relies on at least one method recovering a sufficiently dense candidate graph. In this empirical setting, this condition is not met, so the answer to the second research question is more qualified than the synthetic benchmark alone suggests. The intersection between the LiM and DAGSLAM graphs reduces to LiM's graph itself, with only a single parent, `Prior_Arrest_Episodes_Felony_high`. As a result, the workflow is not informative here, and the output is too sparse to provide meaningful causal insight. We attribute this to the linear SEM assumption being too restrictive for the NIJ dataset.

8.3 Limitations & Further Work

Several factors circumscribe the extent to which conclusions can be drawn from this study and should be kept in mind when interpreting results. We discuss some of them here.

The NIJ dataset was significantly reduced to facilitate tractability, with a lean variable selection chosen to align as closely with criminological theory as feasible. Even so, the Central Eight criminogenic needs in their original form are either absent from the data or only approximated by the closest available proxy variables. The learned causal structures are therefore incomplete and should be interpreted as illustrative rather than definitive, with latent unmeasured confounding likely still present. Future work may benefit from longitudinal data, akin to the approaches taken by some teams in the NIJ challenge, who incorporated PUMA-linked data [42]. The imputation procedure used here may also introduce errors that propagate into the final learned graphs. The extent to which these artefacts affect the recovered structures is difficult to

quantify without a ground truth, contributing a source of uncertainty that incompatibility and edge-stability metrics cannot fully capture.

The dataset is also drawn exclusively from parolees in Georgia observed over the period 2013–2015, so the invariance of any recovered causal structure under different exogenous settings remains undetermined. Considering the transportability of causal structures across geographic settings was one of the motivating objectives of the original project specification, but this could not be met within the available scope. Evaluating recidivism across jurisdictions is challenging because of sensitivity to measurement choices and systematic differences in reporting conventions, policing, courts, and agency practices [61, 62]. Moreover, the heterogeneity of recidivism reporting makes direct quantitative comparison across domains difficult [5]. The dominance of US-centred data in the broader literature compounds this issue, introducing a potential geographical bias. Since one of the motivating appeals of causal discovery is its promise of transportability and domain invariance, the absence of replication across other datasets means this property cannot be empirically verified here. This remains an important direction for future work.

A final major limitation is the violation of assumptions made by several of the algorithms, especially causal sufficiency and, where relevant, linearity. The results presented here therefore do not support any definitive causal claims. Future work may wish to place greater emphasis on algorithms that relax these assumptions, with causal sufficiency appearing especially restrictive in this domain. ProDAG, which was excluded from this study on practical grounds, would also be a valuable addition to future work given its ability to return a distribution over graphs and thereby quantify uncertainty more directly.

9 Conclusion

This thesis set out to answer two questions: how causal discovery algorithms behave on a mixed-type criminal justice dataset, and whether a hybrid multi-algorithm workflow can recover more robust and reliable causal graphs than any single method in isolation.

On the first question, no single algorithm emerged as universally best. All algorithms must balance the trade-off between sensitivity and specificity, and the synthetic benchmark confirmed that performance is domain-dependent, consistent with the findings of Niu et al. [8]. Empirical application to the recidivism setting reinforced this view: linear assumptions appeared too restrictive for the domain, and experiments produced a wide range of graph densities. It was nonetheless encouraging to see repeat hypothetical parents of recidivism across algorithms, including age at release and prior felony convictions, lending credence to such parents.

On the second question, the findings support the view that causal discovery algorithms are best understood as complementary tools whose strengths align with different stages of the discovery process, rather than as direct competitors in a global ranking. Rather than relying on any single method, a practical approach may be to use a more flexible, sensitive algorithm to propose a rich candidate skeleton, apply a conservative optimiser to filter for edges that are reproducible and stable, and then refine orientations using a score-based learner and domain knowledge.

As detailed in Section 8.3, these results are constrained by assumption violations, single jurisdiction scope, and the absence of ground truth. We reiterate that this should only be considered a conceptual heuristic, and used only to inform future studies.

In closing, this work does not purport to deliver a definitive causal graph, nor propose a single winning algorithm. Instead, we aim to demonstrate that these algorithms behave differently and, when used intentionally alongside domain expertise, can be used together to produce better causal structures than any singular algorithm, and expose where current tools are fragile.

A NIJ Dataset Metadata

Column Label	Data type	Categories / Range	Notes
ID	Integer	-	Unique Person ID
Supervision Case Information			
Gender	Categorical	{M, F}	-
Race	Categorical	{Black, White}	-
Age_at_Release	Categorical	{18-22, ... , 48 or older}	7 categories, each representing a 4 year bucket from the ages of 18 through 44.
Residence_PUMA	Categorical	{1, ... , 25}	Residence group at Prison Release categorised according to US Census geographic area
Gang_Affiliated	Binary	-	Verified via investigation.
Supervision_Risk_Score_First	Categorical	{1, ... , 10}	1 denotes lowest risk. Some entries are missing for this field.
Supervision_Level_First	Categorical	{Standard, Specialized, High}	-
Prison Case Information			
Education_Level	Categorical	{Less than HS diploma, High School Diploma, At least some college}	-
Dependents	Categorical	{0, ... , 2 , 3 or more}	-
Prison_Offense	Categorical	{Drug, Violent/Non-Sex, Violent/Sex, Property, Other}	Primary prison conviction offense group. This field contains blanks
Prison_Years	Categorical	{Less than 1 year, 1-2 years, Greater than 2 to 3 years, More than 3 years}	Years in Prison Prior to Parole Release
Prior Georgia Criminal History			
Prior_Arrest_Episodes_Felony	Categorical	{0, ... , 9 , 10 or more}	GCIC arrests where most serious charge was felony.
Prior_Arrest_Episodes_Misd	Categorical	{0, ... , 5 , 6 or more}	GCIC arrests where most serious charge was misdemeanor.
Prior_Arrest_Episodes_Violent	Categorical	{0, ... , 2 , 3 or more}	GCIC arrests where most serious charge was violent.
Prior_Arrest_Episodes_Property	Categorical	{0, ... , 4 , 5 or more}	GCIC arrests where most serious charge was property.
Prior_Arrest_Episodes_Drug	Categorical	{0, ... , 4 , 5 or more}	GCIC arrests where most serious charge was drug related.
Prior_Arrest_Episodes_PPViolationCharges	Categorical	{0, ... , 4 , 5 or more}	Prior GCIC arrests with probation/parole violation charges.
Prior_Arrest_Episodes_DVCharges	Binary	-	Prior GCIC arrests with domestic violence charges.
Prior_Arrest_Episodes_GunCharges	Binary	-	Prior GCIC arrests with gun charges.
Prior Georgia Community Supervision History			

Prior_Conviction_Episodes_Felony	Categorical	{0, ... , 2 , 3 or more}	GCIC convictions where most serious charge was felony.
Prior_Conviction_Episodes_Misd	Categorical	{0, ... , 3 , 4 or more}	GCIC convictions where most serious charge was misdemeanour.
Prior_Conviction_Episodes_Viol	Binary	-	GCIC convictions where most serious charge was violent.
Prior_Conviction_Episodes_Prop	Categorical	{0, ... , 2 , 3 or more}	GCIC convictions where most serious charge was property.
Prior_Conviction_Episodes_Drug	Categorical	{0, ... , 2 or more}	GCIC convictions where most serious charge was a drug charge.
Prior_Conviction_Episodes_PPViolationCharges	Binary	-	-
Prior_Conviction_Episodes_DomesticViolenceCharges	Binary	-	-
Prior_Conviction_Episodes_GunCharges	Binary	-	-
Prior_Revocations_Parole	Binary	-	-
Prior_Revocations_Probation	Binary	-	-
Georgia Board of Pardons and Paroles Conditions of Supervision			
Condition_MH_SA	Binary	-	Parole Release Condition: Mental Health or Substance Abuse Programming
Condition_Cog_Ed	Binary	-	Parole Release Condition: Cognitive Skills or Education Programming
Condition_Other	Binary	-	-
Supervision Activities			
Violations_ElectronicMonitoring	Binary	-	-
Violations_Instruction	Binary	-	-
Violations_FailToReport	Binary	-	-
Violations_MoveWithoutPermission	Binary	-	-
Delinquency_Reports	Categorical	{0, ... , 3 , 4 or more}	-
Program_Attendances	Categorical	{0, ... , 9 , 10 or more}	-
Program_UnexcusedAbsences	Categorical	{0, ... , 2 , 3 or more}	-
Residence_Changes	Categorical	{0, ... , 2 , 3 or more}	-
Avg_Days_per_DrugTest	Float	[0,1]	Blanks in this field
DrugTests_THC_Positive	Float	[0,1]	Blanks present in this field
DrugTests_Cocaine_Positive	Float	[0,1]	Blanks present in this field
DrugTests_Meth_Positive	Float	[0,1]	Blanks present in this field
DrugTests_Other_Positive	Float	[0,1]	Blanks present in this field
Percent_Days_Employed	Float	[0,1]	Blanks present in this field
Jobs_Per_Year	Float	-	Blanks present in this field
Employment_Exempt	Binary	-	-
Recidivism Measures			
Recidivism_Within_3years	Binary	-	-

Recidivism_Arrest_Year1	Binary	-	-
Recidivism_Arrest_Year2	Binary	-	-
Recidivism_Arrest_Year3	Binary	-	-

B Risk assessment tools mapped to the Central Eight

INSTRUMENTS	History of Antisocial Behaviour(*)	Substance Abuse	Familial / Marital Relation-ships	Work/ School	Attitudes	Personality(*)	Associates	Leisure
COMPAS	X	X	X	X	X	X	X	X
OASys	X	X	X	X	X	X	X	X
LSI-R	X	X	X	X	X	X	X	X
LS/CMI	X	X	X	X	X	X	X	X
OGRS3	X							
RISc	X	X	X	X	X	X	X	
PCRA	X	X	X	X	X		X	
ORAS-PAT	X	X		X				
ORAS-CST	X	X	X	X	X	X	X	X
ORAS-CSST	X	X		X			X	
ORAS-PIT	X	X	X	X		X	X	X
ORAS-RT	X	X	X	X	X	X		
HCR-20 V3	X	X	X	X	X	X	X	
Static-99 (Δ)	X		X					
VRAG	X	X	X	X		X		
VRS	X	X	X	X	X	X	X	X
Of a possible 16	16	14	13	14	10	11	11	7

(*) Denotes static factors.
(Δ) Static-99 estimates sexual recidivism risk for adult males, and thus comprises of more targeted factors about the nature of the index crime.

C Causal Pitfalls Ground Truths

C.0.1 Treatment Mediation

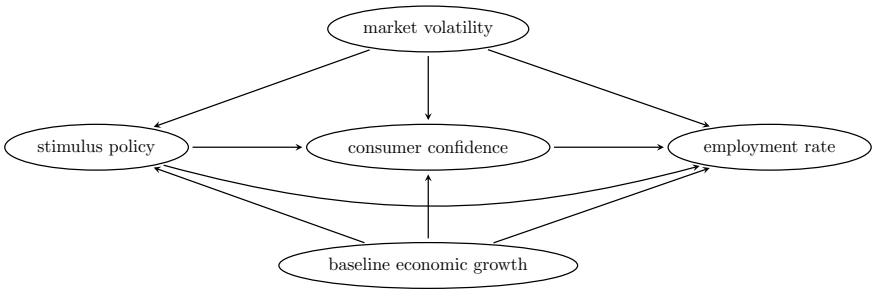


Figure 10: Treatment mediation ground truth.

In this pitfall, we have the treatment (`stimulus_policy`) working through an intermediate variable.

C.0.2 Causal Effect

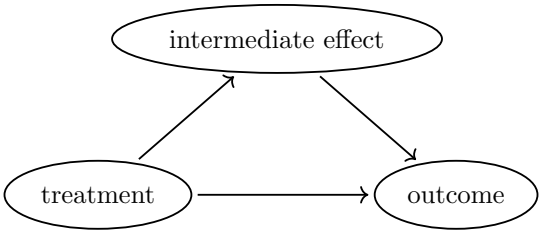


Figure 11: An example of a simple confounded treatment and outcome scenario

Due to algorithmic failures on certain configurations, two datasets were successfully evaluated for this scenario.

C.0.3 Counterfactual Reasoning

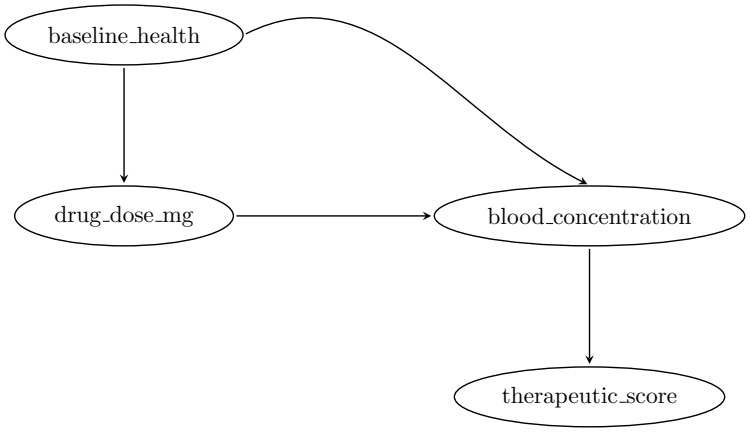


Figure 12: Counterfactual reasoning ground truth

C.0.4 Causal Direction

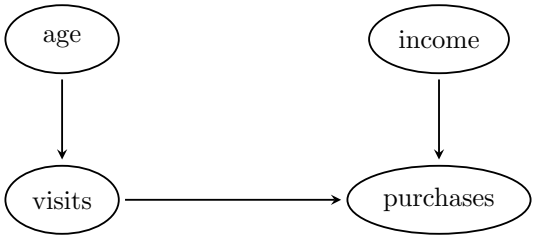


Figure 13: Causal direction scenario ground truth

Due to an algorithmic failure on a certain configuration, four datasets were successfully evaluated for this scenario.

C.0.5 Setting Agnosticism

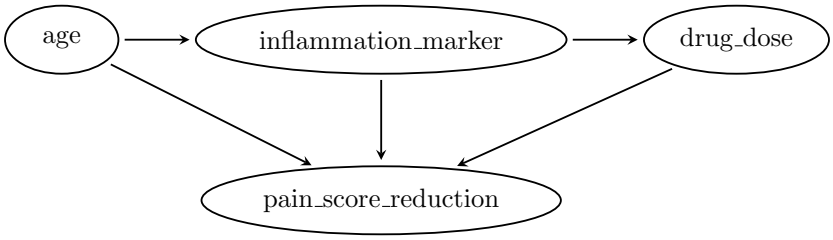


Figure 14: Domain shift scenario ground truth

C.0.6 Confounding Mediators

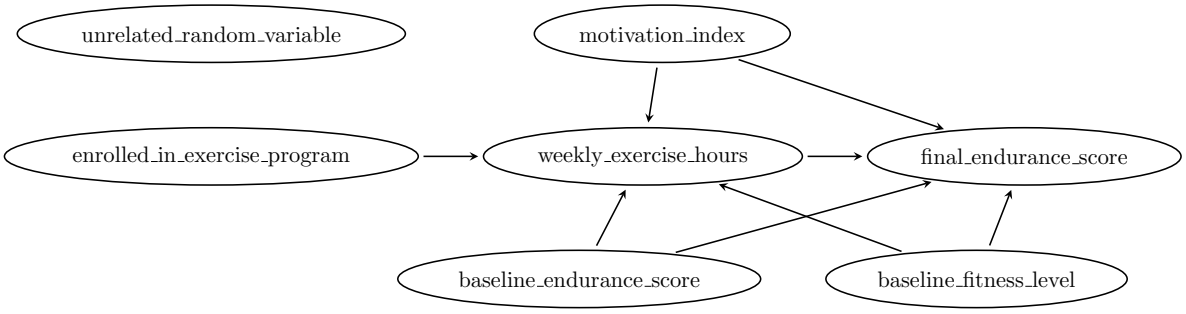


Figure 15: Mediator-outcome confounding ground truth

Three datasets were successfully evaluated for this scenario.

C.0.7 Treatment Effect Modifiers

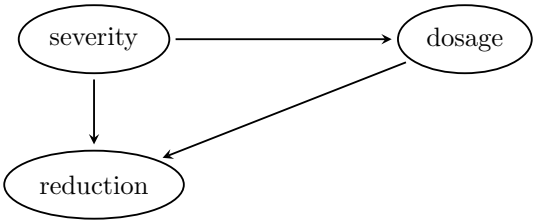


Figure 16: Ground truth of the moderation effect scenario

C.0.8 Markov Equivalences

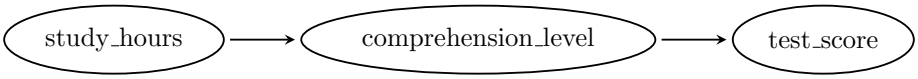


Figure 17: Markov equivalence class scenario

C.0.9 Necessity and Sufficiency

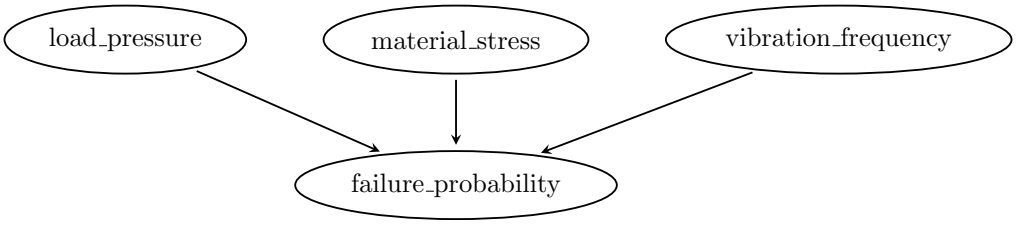


Figure 18: Necessity and sufficiency ground truth

C.0.10 Sequences of Mediators

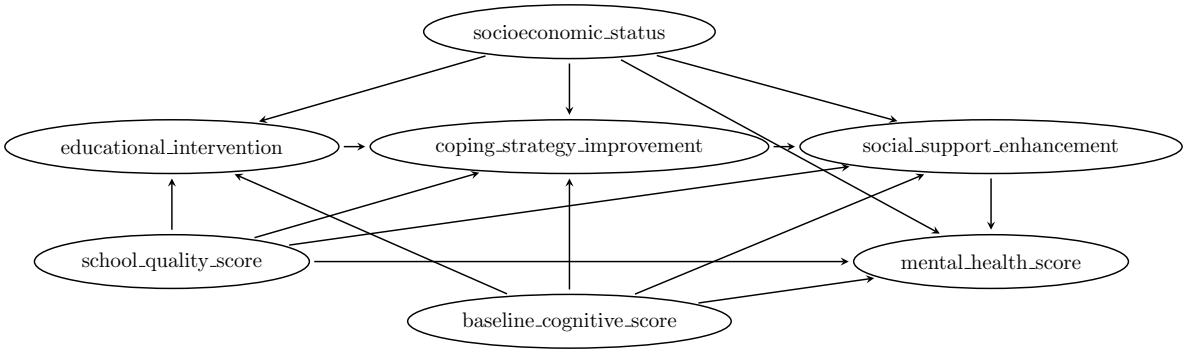


Figure 19: Sequential mediator ground truth

D Datasets run for each Causal Pitfall scenario

Causal Pitfall Scenario	Total Datasets	Datasets Successfully Evaluated
Treatment Mediation	5	5
Causal Effect	5	2
Counterfactual	5	5
Causal Direction	5	4
Domain Shift	5	5
Mediation Outcome Confounder	5	3
Moderation Effect	5	4
Markov Equivalence	5	5
Necessity Sufficiency	5	5
Sequential Mediator	5	5

Table 14: Table collating the number of datasets present and successfully evaluated per causal pitfall scenario. Discrepancies reflect algorithmic failures on certain data configurations.
** A dataset is counted as successfully evaluated if all algorithms produced a valid output for that configuration.*

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